Question Paper ID: 258354

Course Code: AI365TDD

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

USN 1 R V 2 3 A I 0 7 2

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)
R V Vidyanikethan Post
Mysuru Road, Bengaluru - 560059

VI Semester B.E. Regular / Supplementary Examination July / August 2026

Generative Artificial Intelligence

Time : 3 Hours

Instructions to the students **Maximum Marks : 100**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Define latent space.	02	2	2
1.2	What does the encoder produce in a Variational Autoencoder instead of a fixed latent vector?	02	1	1
1.3	Along with mathematical notation, define a Generative Adversarial Network (GAN).	02	1	1
1.4	How does a CycleGAN perform image-to-image translation without paired datasets?	02	2	3
1.5	Write the total loss equation used in a Variational Autoencoder.	02	2	2
1.6	Describe the primary function of the Generator in a Generative Adversarial Network (GAN).	02	1	2
1.7	In brief describe the Reverse Diffusion Process.	02	2	1
1.8	Explain an Energy-Based Model (EBM).	02	2	2
1.9	State any two ethical AI design principles used in Generative AI.	02	2	2
1.10	List any two deployment challenges associated with Generative AI systems.	02	2	2

Part B

Question No	Question	M	CO	BT
2a	An autoencoder is used to compress image data through its encoder. Consider an input image of size $28 \times 28 \times 1$. The encoder consists of convolutional layers with the following parameters: ☐ Layer 1: 16 filters, kernel size = 3×3, stride = 2, padding = 1 ☐ Layer 2: 32 filters, kernel size = 3×3, stride = 2, padding = 1 Write a Python program (using Py Torch or TensorFlow) to implement the encoder.	08	4	6
2b	Construct a model that uses self-attention, masking, and positional information to encode an input sequence and decode an output sequence. Draw the architecture and describe each stage.	08	2	4
3a	Discuss the role of Autoencoders as denoising models. Analyze the impact of selecting very small and very large latent spaces.	08	2	1
3b	Demonstrate the latent space sampling process in a Variational Autoencoder. Illustrate the mathematical formulation and workflow.	08	2	2
	OR			
4a	Discuss Autoencoders as denoising models. Analyze how the choice of latent space dimension influences reconstruction quality and model generalization.	08	3	2
4b	Illustrate the process of latent space sampling in a Variational Autoencoder. Derive the sampling equation and discuss its significance in data generation.	08	3	4
5a	Explain the roles of the generator and discriminator in a Generative Adversarial Network (GAN) and how adversarial training improves image generation quality.	08	2	2
5b	What role does the PatchGAN discriminator play in CycleGAN training and image quality? Explain in detail.	08	3	2
	OR			
6a	An image generated by Neural Style Transfer preserves style but loses object structure. Explain how the loss functions can be adjusted to improve the output.	08	3	3

6b	Design and compare the ResNet and U-Net generator architectures used in CycleGAN. Explain how each architecture affects the translation quality, particularly in terms of content preservation and style application.	08	3	4
7a	Explain the architecture and working principle of a Denoising Diffusion Model (DDM) with a neat diagram.	08	3	2
7b	Compare the Forward Diffusion Process and the Reverse Diffusion Process with suitable examples.	08	2	3
	OR			
8a	Explain the MNIST dataset and discuss its significance in training Energy-Based Models.	08	2	2
8b	Illustrate the sampling process in an Energy-Based Model using Langevin Dynamics.	08	2	3
9a	Describe the three bias mitigation strategies used in Generative AI with suitable examples.	08	2	2
9b	Illustrate the role of monitoring and feedback loops in the responsible deployment of Generative AI systems.	08	1	1
	OR			
10a	Critically analyze the impact of algorithmic, data, and societal biases on Generative AI systems. Illustrate your answer with suitable real-world examples and suggest appropriate mitigation strategies.	08	3	4
10b	Compare the fairness metrics used in Generative AI and evaluate their suitability for different real-world applications.	08	2	2